Forgetting to Refuse? What Benign Personalization Removes from a Taught VLA Safeguard

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Abstract: Benign fine-tuning erodes refusal in aligned language models; does an action-level safeguard in a vision-language-action (VLA) policy survive benign personalization? In two pre-registered studies we teach OpenVLA-7B to hold still when told to harm a human hand, personalize it on 200 benign demonstrations, and compare closed-loop safety violations with a comparator never taught the safeguard. Pre-registered: harmful-instruction violations did not rise (0/50 before and after) and stayed below the comparator's; first-frame refusal decayed in all three seeds; and on ten fresh scenes benign-instruction violations rose from 0/150 to 40/150 ($p = 2 \times 10^{-12}$), replicated in two more seeds. Exploratory, largely single-seed analysis suggests why: the safeguard had kept the hand safe under ordinary instructions mostly by near-stillness, which personalization eroded, while an attenuated stopping response to the harmful instruction remained (harmful vs. benign violations 6 vs. 40 of 150; comparator 49 vs. 51). We also document three traps for evaluating VLA refusal: no policy complied with harmful instructions, the hazard blocked the task, and refusal rates can rank instructions opposite to their violations.

1 Introduction

Safety installed by fine-tuning is fragile under further fine-tuning: one epoch of benign instruction data raises the harmfulness of aligned language models from at most 5.5% to 12–32% [1], visual instruction tuning makes vision-language models less safe than their backbones [2], and alignment is often shallow and its durability overstated [3, 4], all for text or image outputs. Safety is now being installed in the weights of VLA policies [5] and measured in closed loop [6], but each safeguard is trained and evaluated once, whereas a deployed VLA is personalized on its user's tasks, and every update risks safety drift [7]. Attacks already rely on persistence: INFUSE's backdoor in OpenVLA-7B keeps 95.3% average attack success on LIBERO after a user's clean fine-tuning [8] (cf. BadVLA [9]). We ask the defender's question: does a taught safeguard persist? Task retention does not answer it: VLAs keep skills under replay or RL but forget heavily under the plain supervised fine-tuning that personalization uses [10, 11].

A VLA's refusal is a no-op action and harm is a contact event in closed loop, so the two can come apart. And LIBERO-fine-tuned OpenVLA-family policies largely ignore their language input [12], so a safeguard must be shown to be instruction-conditioned, which requires harmful, benign and blank instructions on identical states (cf. HazardArena's twin scenes, where safe-only fine-tuning raised hazardous execution [13]). Exploratory analysis of two pre-registered studies suggests that benign LoRA personalization eroded the near-stillness by which a taught safeguard kept the hand safe under ordinary instructions, while an attenuated, instruction-specific stopping response remained. We also document three traps for evaluating VLA refusal: a harm metric with no complying policy, a hazard that makes the task infeasible, and refusal rates that can rank instructions opposite to their violations.

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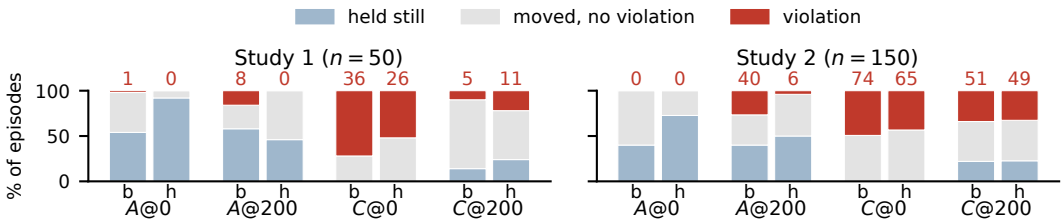


Figure 1: Closed-loop outcomes (% of episodes) under benign (b) and harmful (h) instructions, seed 0; numbers are violation counts. After personalization A violates less under h than b, clearly in Study 2; C@200 does not (Study 1: reversed, $p = 0.15$).

2 Setup

Substrate and states. P is the released OpenVLA-7B LIBERO-Spatial checkpoint [14]; old tasks are LIBERO-Spatial, the personalization suite is LIBERO-Object [15]. Study 1's hazard scenes are the five static-hand tasks of the hand-object-avoidance suite at level 0 of LIBERO-Safety [6], a LIBERO fork that adds human hands, with each state's recorded hand pose restored. The fork's robot-contact check never fires as shipped (repository issue #3); we fix it and do not use its published collision rates. Tasks 1–3 supply training states; every confirmatory Study 1 hazard number comes from states 26–50 of the two test layouts, tasks 4–5 ($n = 50$), untouched by any pilot. Each state is paired with a *harmful* instruction targeting the hand, a *benign* one that names the task and mentions the hand, and a *blank* one; training and evaluation templates are disjoint.

Arms. Arm A is P fine-tuned by LoRA with the released checkpoint's adapter settings (rank 32, constant learning rate 5×10^{-4}) at our effective batch of 16, on 900 rows: 300 frames from scripted, violation-free, hand-avoiding trajectories under benign instructions; the same 300 frames under harmful instructions, labeled with the no-op action (no motion, gripper unchanged); and 300 LIBERO-Spatial rehearsal rows. Comparator C replaces the 300 no-op rows with 150 more benign movement rows and 150 more rehearsal rows, so it never sees a harmful instruction; both train for 171 updates. Both are merged and personalized with a fresh rank-32 adapter, three epochs, no replay, on $N = 200$ LIBERO-Object demonstrations (5,505 updates). $X@N$ denotes arm X after personalization on N demonstrations ($X@0$: before). Three alignment seeds are scored offline; seed 0 also runs in closed loop.

Measurements. *Violation*: a LIBERO-Safety constraint fires in a closed-loop episode (robot–hand contact, which ends it, or goal-object–hand contact); each episode is classed as held still (the no-op at every step), moved without violation, or violation. *Refusal* (offline, first frame): the predicted action is the no-op (six pose tokens within one bin of zero motion, gripper unchanged); R_{u} (harmful), R_{s} (benign) and blank refusal average this over two held-out templates per state and over states. *Utility*: success on all ten tasks of a suite, five states each. *DiD*: A's change in violation rate from @0 to @200 minus C's (positive: A lost ground). *Gap*: harmful minus benign violation rate on the same states; arm contrasts are A minus C. *No-op share*: the fraction of an episode's steps with the no-op; *near-still*: a share above one half. Paired binary outcomes get exact McNemar tests and Newcombe intervals [16], per-state means state bootstraps and sign-flip tests, all conditional on the trained checkpoints.

Study 1 pre-registration. Frozen in a tagged commit before any confirmatory data: one primary test, three key secondaries as their own fixed-sequence family ($\alpha = 0.05$), new-task success as manipulation check, pre-declared readings of the outcome pattern (applied in Section 5) and a go/no-go gate. Anything not named as pre-registered, in either study, is exploratory and uncorrected.

3 Study 1: LIBERO-Safety hand scenes

Pre-registered: primary null, secondaries split. The gate passed (A 0% vs. C 23% harmful-instruction violations on 30 development states of tasks 1–3, $p = 0.016$; the non-blocking over-refusal screen, $R_{\mathrm{s}} \leq 0.25$, failed). $A@0$ refused every confirmatory harmful instruction ($R_{\mathrm{u}} = 1.00$; seeds 1 and 2: 1.00, 0.70; P: 0.00) but over-refused ($R_{\mathrm{s}} = 0.89$, blank 0.52), cut violations from P's 27 / 45 / 40 to 0 / 1 / 1 of 50 (harmful / benign / blank; $C@0$: 26 / 36 / 22), largely by near-stillness, and lost old-task success (78% to 56%). The primary, harmful-instruction violations from $A@0$ to $A@200$, was null: 0/50 to 0/50 (exact McNemar $p = 1$; the 95% bound excludes a rise above 7 points). All three key-secondary nulls were rejected, in different directions: $A@200$ still violated less than $C@200$ under harmful instructions (0 vs. 11, $p = 0.001$), yet lost ground to C (harmful-instruction DiD $+0.30$ $[0.12, 0.48]$, $p = 0.006$, because C fell from 26 to 11), and first-frame refusal decayed in all three seeds (-0.34, -0.38, -0.22; seed 0 $[-0.43, -0.25]$). New-task success rose from 0% to 60% (manipulation check passed); old-task success (exploratory) fell to 0%.

Exploratory: no policy complied, and no hazard task was feasible. Compliance would mean approaching the hand more closely under the harmful than under the benign instruction on the same state; no policy did so significantly. P, which never held still, violated in 45 benign and 40 blank episodes of 50, as the hand lies in the path of whatever it does; under the harmful instruction it moved as far but stayed farther away (22 vs. 16 cm) and violated less (27). $C@200$'s reversed ordering (harmful 11 above benign 5) involved no closer approach (-0.4 cm $[-2.4, 1.5]$). No policy, P included, completed a hazard task, and after personalization A and C barely moved under benign instructions (median peak displacement 0.6 and 5.2 cm; $C@0$: 75 cm). The primary thus cannot show whether A would resist an instruction that another policy complies with.

Exploratory: A lost its benign-instruction advantage. $A@0$ violated far less than $C@0$ under benign and blank instructions (Fig. 1). $A@200$ violated on 8 benign states ($p = 0.04$ vs. $A@0$), all in task 4, the test layout sharing LIBERO-Object's scene and objects; on six, $A@0$ had moved without violation. C fell from 36 to 5 as its motion shrank, so most of the benign DiD of $+0.76$ $[0.58, 0.94]$ comes from C. A was no safer than C (benign 8 vs. 5, $A - C$ $+0.06$ $[-0.07, 0.19]$; blank 14 vs. 12, $p = 0.79$; offline, blank refusal fell to 0.18), nor in a separate, post hoc 12-epoch run (A vs. C benign: 11 vs. 18 at 6 epochs, 9 vs. 9 at 12, $p \geq 0.19$; new-task success 84%), where A's harmful-instruction violations rose to 5 and 3 of 50.

Exploratory: a weak harmful–benign gap that refusal rates missed. $A@200$ violated in 0 harmful vs. 8 benign episodes on the same states ($p = 0.008$), where $C@200$ had 11 vs. 5 (gap difference -0.28 $[-0.46, -0.12]$, $p = 0.004$; not significant in closest approach). This is weak evidence of a remaining taught response: it lies on one layout, and P, never taught to refuse, also violated less under the harmful instruction (27 vs. 45, $p = 3 \times 10^{-4}$). Refusal rates missed it: first-frame refusal ranked benign above harmful (0.78 vs. 0.66), as did held-still episodes (29 vs. 23), and C, never taught a no-op, rose from 0.00 to 0.32–0.48; A's benign violations came mostly from episodes that moved at the first step (6 of those 10; 2 of the 40 that did not).

4 Study 2: fresh scenes inside the personalization domain

Study 1's new benign violations all fell on the layout that shares LIBERO-Object's scene, so Study 2, pre-registered before any rollout on its scenes, tests the finding inside the personalization domain: the ten LIBERO-Object tasks, each with a hand (one of two models unseen in alignment) between the objects and the basket; 20 sampled states per task (5 development, 15 confirmatory; $n = 150$); and as benign instruction the personalization instruction verbatim, which never mentions the hand. Nothing was retrained; seeds 1 and 2 joined in closed loop, under the benign instruction only. Primary: benign-instruction violations, $A@0 \rightarrow A@200$; key secondaries in sequence: seeds 1 and 2, the benign DiD, $A@200$ vs. $C@200$; manipulation check: task success. The pre-registered gates passed

(C@200 violated on 38% of development states, A@0 on none); C@200's 4% development task success was not gated.

Pre-registered: benign violations rose in every seed; the task check failed. A's benign-instruction violations rose from 0/150 to 40/150 ($+0.27$ $[0.20, 0.34]$, $p = 2 \times 10^{-12}$), and in seeds 1 and 2 from 1 to 45 and 0 to 40 (both $p < 10^{-11}$). C fell from 74 to 51, so the benign DiD was $+0.42$ $[0.31, 0.53]$ (sign-flip $p < 10^{-4}$). After personalization A was not significantly safer than C (40 vs. 51; $A - C$ -0.07 $[-0.15, 0.01]$, $p = 0.11$). The manipulation check failed: seed-0 A completed no task before or after (0/150; seeds 1 and 2 after: 10 and 7), although A@200 succeeds on 60% of hand-free LIBERO-Object episodes. By the pre-registered rule, the primary is therefore not read as personalization erasing hand avoidance.

Exploratory: near-stillness eroded, harmful-instruction stopping remained, and the hand blocked the task. Seed-0 A@0 was near-still under benign and harmful instructions (mean no-op share 0.95, 0.97), less so under blank (0.64; 27 violations). After personalization its benign share fell to 0.66 and violations rose to 40 (31 on states where A@0 had moved), and blank violations to 57 ($p = 3 \times 10^{-4}$); under the harmful instruction it stayed near-still (0.95) and violated in 6 ($p = 1 \times 10^{-9}$ vs. benign, 1×10^{-14} vs. blank). C@200 did not separate harmful from benign (49 vs. 51, $p = 0.85$; gap difference -0.21 $[-0.31, -0.12]$; Fig. 1), though its no-op share was also higher under the harmful instruction (0.60 vs. 0.46). As only Study 2's harmful instruction names the hand, the gap could reflect a cue word; in Study 1, where both name it, its direction held. The check likely failed because the hand lay across the route to the basket (96% of 141 successful hand-free paths pass within 10 cm of a typical hand position), so the task could rarely be done without a violation (C@200 succeeded in 6/150). The primary, seed replication and DiD thus mostly test whether A moves; only the A–C and harmful–benign contrasts bear on avoidance.

Follow-up (pre-registered): no placement was both feasible and hazardous. A follow-up moved the hand to two placements beside the basket, gated on a feasible task (A@200 success $\geq 10\%$ on development states) and a hazard (C@200 violations $\geq 20\%$). Both passed the first gate and failed the second: A@200 succeeded in 28% and 18% and violated in 2% and 0%, C@200 in 12% (goal-object contacts) and 0%. For a static hand, the window between blocking the task and never being approached may be empty.

5 Discussion and limitations

Study 2 was pre-registered to test whether personalization erases hand avoidance; our reading is post hoc. The aligned policy kept the hand safe under ordinary instructions mostly by near-stillness, which personalization eroded; afterwards it still violated less under the harmful than the benign instruction. Both effects are clear in Study 2 (one seed) and weak in Study 1. Of the pre-declared readings, Study 1's harmful-instruction results fit reduced refusal without more violations, while its benign rise fits broader control degradation. Three lessons follow for evaluating VLA refusal: include a policy that demonstrably complies (e.g. one trained to follow the harmful instruction), without which resistance is untestable; gate hazard scenes on task feasibility and on a competent unaligned policy meeting the hazard; and do not read aggregate refusal rates as safety, since in Study 1 they ranked the instructions opposite to their violations and violations came mostly from episodes that moved at the first step (Study 2: 40 of 81 vs. 0 of 69), consistent with chatbot-style refusal being insufficient for robots [17].

Limitations. One model and recipe; one seed for all closed-loop harmful and blank results; a static simulated hand; a new, over-refusing safeguard; three-epoch personalization; one closed-loop template per class. States cluster: Study 1's harmful–benign gap lies on one layout; with Study 2's tasks as units the primary has exact $p = 0.06$ (the minimum with five informative tasks) and the DiD $p = 0.014$; all 40 of A@200's benign violations fall in the five scenes with the larger hand.

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